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No. of printed pages : 02

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SARDAR PATEL UNIVERSITY

S.Y.B.C.A. 4th Sem [NEP] Examination- April 2025

US04MABCA02 || Operating System



Date: 03/04/2024 Thursday

Time: 02:00pm to 04:30pm

Maximum Marks: 50

Q.1

Answer the following questions.

[10]

- 1 Time- Slice Scheduling algorithm is also called _____.
(a) Priority (b) Round-Robin (c) FCFS (d) SJF
- 2 In Layered Approach, bottom layered being _____.
(a) System Program (b) User Interface (c) Hardware (d) Application Program
- 3 A Program in execution is known as _____.
(a) Process (b) Files (c) Job (d) None of these
- 4 Technique use to solve problem of external fragmentation is known as _____.
(a) Process (b) Compaction (c) Aging (d) None of these.
- 5 _____scheduling algorithm gives minimum Page Faults.
(a) FIFO (b) LRU (c) Optimal (d) None of these.
- 6 Valid-Invalid bit scheme in page table is used in _____.
(a) Demand paging (b) Fragmentation (c) Compaction (d) None of these
- 7 The concept in which output depends on sequence of process execution is known as _____.
(a) output-condition (b) Race-Condition (c) Run (d) None of these.
- 8 _____is the solution of the bounded buffer.
(a) Data consistency (b) Virtual memory (c) Shared-Memory (d) None of these
- 9 Which command is used to remove directories?
(a) rmdir (b) rm (c) deldir (d) dirdel
- 10 To evaluate conditional expression in Linux command is used.
(a) expr (b) testexpr (c) eval (d) test

[P.T.O.]

Q.2 Explain Process Control Block with information stored in it. **[10]**

OR

Q.2 Discuss SJF and Round-Robin scheduling algorithm with suitable example. **[10]**

Q.3 Discuss first-fit, best-fit and worst-fit memory allocation algorithms through example and write merits and demerits of each one. **[10]**

OR

Q.3 List different page Replacement algorithms and Explain Optimal Page Replacement Algorithm and FIFO page replacement algorithm with example. **[10]**

Q.4 Explain Deadlock through appropriate example. Explain all necessary conditions for occurrence of deadlock. **[10]**

OR

Q.4 Explain Critical-section problem in detail. Discuss Peterson's algorithm (Algorithm-3) for two-process solution. **[10]**

Q.5 Discuss all conditional statements (Control Structure) using appropriate example. **[10]**

OR

Q.5 Discuss echo, cd, man, expr and grep command with use, syntax and suitable example. **[10]**

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